O–G–491a
FEBRUARY 17, 1955

SUPERSEDING
Int. Fed. Spec. O–G–00491 (Army–Ord.)
June 2, 1953 and
Fed. Spec. O–G–491
February 3, 1931

FEDERAL SPECIFICATION

GLYCERIN (GLYCEROL)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. CLASSIFICATION

1.1 Grades.—Glycerin covered by this specification shall be of the following grades, as specified (see 6.2):

Grade A.—U. S. P. glycerin.

Grade B.—High-gravity ("Dynamite") glycerin.

Grade C.—Yellow distilled glycerin.

2. APPLICABLE SPECIFICATIONS AND OTHER PUBLICATIONS

2.1 Specifications.—There are no other specifications applicable to this specification except those referenced in section 7 by individual departments.

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

2.2 Other publications.—The following publication, of the issue in effect on date of invitation for bids, forms a part of this specification:

The Pharmacopoeia of the United States.

(Copies of the Pharmacopoeia may be obtained from the U. S. Pharmacopoeial Committee, 46 Park Avenue, New York 16, New York.)

3. REQUIREMENTS

3.1 Physical and chemical properties.—The glycerin shall conform to the corresponding chemical and physical requirements specified hereunder.

3.1.1 *Grade A glycerin.*—The glycerin shall conform in all respects to the requirements specified in the Pharmacopoeia of the United States.

3.1.2 *Grade B glycerin.*

3.1.2.1 *Appearance.*—The glycerin shall be clear and free of suspended matter.

3.1.2.2 *Odor.*—The odor of the glycerin shall be slight, or when so specified it shall conform to the odor of a sample which has been mutually agreed upon by the procuring agency and the contractor (see 6.2).

3.1.2.3 *Color.*—The color of the glycerin shall be not darker than that of the color standard (see 4.3.1).

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3.1.2.4 *Specific gravity.*—The specific gravity of the glycerin at 15.5°/15.5°C. shall be not lower than 1.2620 (see 4.3.2).

3.1.2.5 *Acidity or alkalinity.*—A 50-ml. portion of the glycerin shall require not more than 0.30 ml. of normal hydrochloric acid solution, or not more than 0.30 ml. of normal sodium hydroxide solution, for neutralization (see 4.3.3).

3.1.2.6 *Ash.* — The ash content of the glycerin shall not exceed 0.10 percent by weight (see 4.3.4).

3.1.2.7 *Total chlorides (including chlorinated compounds).*—The total chlorides content of the glycerin, expressed in terms of chlorine concentration, shall not exceed 30 parts of chlorine per million (see 4.3.5).

3.1.2.8 *Saponification equivalent.* — The saponification equivalent, calculated as sodium oxide (Na_2O), shall not exceed 0.05 percent (see 4.3.6).

3.1.2.9 *Silver nitrate precipitation.*—No flocculent precipitate shall form within 10 minutes following the addition of silver nitrate solution to the glycerin (see 4.3.7).

3.1.3 *Grade C glycerin.*

3.1.3.1 *Appearance.* — The glycerin shall be clear and free of suspended matter.

3.1.3.2 *Odor.*—The odor of the glycerin shall not be disagreeable, or when so specified it shall conform to the odor of a sample mutually agreed upon by the procuring agency and the contractor (see 6.2).

3.1.3.3 *Color.* — The color of the glycerin shall be not darker than that of the color standard (see 4.3.1).

3.1.3.4 *Specific gravity.*—The specific gravity of the glycerin at 15.5°/15.5°C. shall be not lower than 1.2550 (see 4.3.2).

3.1.3.5 *Acidity or alkalinity.*—A 50-ml. portion of glycerin shall not require more than 0.30 ml. of normal hydrochloric acid solution, or more than 0.30 ml. of normal sodium hydroxide solution, for neutralization (see 4.3.3).

3.1.3.6 *Ash.*—The ash content of the glycerin shall not exceed 0.02 percent by weight (see 4.3.4).

3.1.3.7 *Total chlorides (including chlorinated compounds).*—The total chlorides content of the glycerin, expressed in terms of chlorine concentration, shall not exceed 30 parts chlorine per million (see 4.3.5).

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Lot.—A lot shall consist of all glycerin of the same grade produced in the same batch or blending operation. In the event of a continuous process a lot shall consist of all glycerin subjected to the same processing operations and conditions.

4.2 Sampling.—The sampling of the glycerin in a lot for test purposes shall be performed as specified hereinafter. (Note.—If possible sampling should be made in clear, dry weather as glycerin is very hygroscopic.)

4.2.1 *Small (1-pint or 1-quart) containers.*—Four 1-pint containers, or two 1-quart containers, shall be taken at random from the lot by the inspector, and forwarded to the testing laboratory designated by the procuring agency.

4.2.2 *Large (1-gallon or 5-gallon) containers.*—Two of the containers in the lot shall be taken at random by the inspector. The material in each of the two containers shall be mixed to a uniform consistency throughout, and a 16-ounce sample taken from each container. Each of the two samples shall be placed in separate clean, airtight glass or metal containers, and labeled to show the name of the material, grade, manufacturer, plant, contract or order number, lot number, and lot size. All the tests specified herein shall be made on each of the two samples.

4.2.3 *Tank cars.*—A 1-quart sample shall be taken from the tank car by means of a clean,

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weighted, narrow-neck glass bottle. The unstoppered bottle shall be rinsed with the material being sampled, lowered to the bottom of the tank by means of a cord, chain, or rod, and immediately withdrawn to the surface. In order to obtain a representative cross-section of the material in the tank car the speed of lowering and raising of the bottle shall be uniform, and so regulated that the bottle is just filled as it reaches the surface of the liquid. The sample shall be placed in a clean, airtight glass or metal container, labeled to show the name of the material, manufacturer, plant, contract or order number, lot number, and lot size. All the tests specified herein shall be made on this composite sample representative of the lot.

4.3 Test procedures.

4.3.1 *Color.*

4.3.1.1 *Preparation of standard color solution.*—Dissolve reagent-grade ferric chloride in a solution consisting of 25 volumes of hydrochloric acid and 975 volumes of distilled water. Adjust the strength of the resulting solution so that the ferric iron content corresponds to 45.05 milligrams of $FeCl_3 \cdot 6H_2O$ per milliliter of solution.

4.3.1.2 *Procedure.* — Fill a 50-ml. Nessler tube to the mark with the glycerin under test. Fill another 50-ml. Nessler tube to the mark with the standard color solution. Compare the color of the glycerin with that of the standard by viewing through the contents of both tubes downward against a white background.

4.3.2 *Specific gravity.* — Determine the specific gravity of the glycerin at 15.5°C., in comparison with that of water at the same temperature, by any method that is accurate to within 0.0002 unit.

4.3.3 *Acidity or alkalinity.*

4.3.3.1 *Reagents.*

4.3.3.1.1 *Standard sodium hydroxide solution (approximately 0.5N).* — Prepare a saturated solution of NaOH in freshly boiled distilled water in a stoppered flask, and allow it to stand overnight. Transfer a 35-ml. portion of the clear solution to a 1,000-ml. volumetric flask. Dilute to the mark with recently boiled distilled water, mix thoroughly, cool to room temperature, and standardize against National Bureau of Standards standard acid potassium phthalate (see 6.3), using phenolphthalein indicator. Dilute if desired, restandardize, and protect the solution from carbon dioxide.

4.3.3.1.2 *Standard hydrochloric acid solution (approximately 0.5N).*—Add distilled water to pure hydrochloric acid (specific gravity 1.19), until the specific gravity of the solution is approximately 1.020. Transfer 500 ml. of this solution to a volumetric 1,000-ml. flask, dilute to the mark with distilled water, and mix thoroughly. Standardize the solution against the standardized NaOH solution (see 4.3.3.1.1.), using phenolphthalein indicator, or by suitable gravimetric method.

4.3.3.1.3 *Phenolphthalein indicator.* — Dissolve pure phenolphthalein in 50-percent ethyl alcohol in the proportion of 5 grams phenolphthalein per 1,000 ml. of solution.

4.3.3.2 *Procedure.* — Measure a 50-ml. portion of the glycerin sample in a graduated cylinder, and pour it into a 200-ml. Erlenmeyer flask. Rinse the graduated cylinder with 100-ml. of freshly boiled and cooled (carbon dioxide-free) distilled water that has been rendered exactly neutral to phenolphthalein indicator, and pour the rinsings into the flask. Titrate the glycerin solution to neutrality with 0.5*N* HCl solution or with 0.5*N* NaOH solution, as required, using phenolphthalein indicator. Multiply the number of milliliters of reagent solution required by the normality of the solution used, and report the product as milliliters of normal solution required for neutralization of 50-ml. of the sample.

4.3.4 *Ash.*—Heat approximately 50 grams of the sample in a tared dish over a free flame until the vapor continues to burn after removing the heat source. Carefully protect the operation

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from drafts. When the combustion stops, ignite the residue at low red heat until the carbonaceous matter is consumed. Cool in a desiccator, and weigh. Calculate the increase in weight of the dish to percentage of ash content.

4.3.5 *Total chlorides.* — Transfer a 5-gram portion of the sample, accurately weighed, to a dry 100-ml. flask equipped with a standard-taper ground-glass joint. Add 15 ml. of morpholine, attach the flask to a condenser, and reflux the contents gently for 3 hours. Rinse the condenser with 10 ml. of water, and remove the flask from the source of heat. Allow the flask and contents to cool to room temperature. Make the mixture acid to litmus with concentrated nitric acid. Transfer the solution to a 50-ml. Nessler tube, and add 1 ml. of 0.05*N* silver nitrate solution. Dilute with water to the mark and mix thoroughly. The turbidity shall be not greater than that produced in a blank consisting of 15 ml. of morpholine, 0.2 ml. of 0.020*N* hydrochloric acid solution, and distilled water to fill a 50-ml. Nessler tube to the mark. (The blank corresponds to a chlorine concentration of 30 parts per million).

4.3.6 *Saponification equivalent (applicable to grade B only).*—Transfer a 25-gram portion of the sample, accurately weighed, to a 250-ml. Erlenmeyer flask. Add 100 ml. of distilled water which had been previously boiled to expel carbon dioxide. By means of a pipette transfer 10 ml. of approximately 0.33*N* sodium hydroxide solution to the flask. Attach the flask to a condenser and reflux the contents gently for 1 hour. Remove the flask and contents from the source of heat, allow the contents to cool to room temperature, and titrate the excess sodium hydroxide with 0.1*N* hydrochloric acid, using phenolphthalein as indicator. Make a blank determination, using 100 ml. of the carbon dioxide-free water, and 10 ml. of sodium hydroxide, and heating and titrating in the same manner. Calculate the saponification equivalent, as percent sodium oxide, as follows:

$$\text{Saponification equivalent (percent } Na_2O) = \frac{3.10\ (A-B)\ N}{W}$$

where:

A = ml. of hydrochloric acid solution used to titrate the blank.

B = ml. of hydrochloric acid solution used to titrate the sample.

N = Normality of the hydrochloric acid solution.

W = weight of the sample.

4.3.7 *Silver nitrate precipitation (applicable to grade B only).*—By means of a pipette transfer a 10-ml. portion of the sample to a 25-ml. glass-stoppered cylinder, and add exactly 10 ml. of 10-percent silver nitrate solution. Mix the solution thoroughly by repeated inversion of the cylinder. (Note.—The mixing operation should not be so vigorous as to produce a froth. Both the addition of the silver nitrate solution and the mixing operation should be done in subdued light.) Allow the solution to stand for 10 minutes, and then examine the contents of the cylinder for the formation of a flocculent precipitate. If a flocculent precipitate (not to be confused with a finely divided precipitate of silver) is formed, the glycerin shall be considered to have failed this test.

4.4 Resubmission and retest.—If the composite sample, or any primary sample subjected to test, fails to pass the required tests, the lot shall be rejected. The contractor shall have the option of having a partial or complete analysis made on samples taken from all or any of the containers in the lot at no expense to the Government. The contractor may then remove the defective portion of the lot, and resubmit the lot for acceptance provided complete replacement of the defective portion can be made to the satisfaction of the inspector. The resubmitted lot shall be accepted, provided that new samples, selected in accordance with 4.2, pass all the tests required by this specification, and that 3 additional primary samples (except in the case of tank car lots) pass any tests failed by the material on original submission.

5. PREPARATION FOR DELIVERY

5.1 Packaging. — Unless otherwise specified, commercial packages are acceptable under this specification.

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5.2 Packing.—Unless otherwise specified, the glycerin shall be shipped in standard commercial containers, so constructed as to insure acceptance by common or other carriers, for safe transportation, at the lowest rate of the carrier to the point of delivery.

5.3 Marking. — Unless otherwise specified, shipping containers shall be marked with the name of the material, grade, and the quantity contained therein as defined by the contract or order under which the shipment is made, the name of the contractor, and the number of the contract or order.

6. NOTES

6.1 Intended use.—The glycerin covered by this specification is intended for the following uses:

(a) Grade A glycerin is known in the trade as "U. S. P.-grade" glycerin.

(b) Grade B glycerin is used in the manufacture of explosives, and in other industrial applications for which high-gravity glycerin is required.

(c) Grade C glycerin is used in industrial applications other than for the manufacture of explosives.

6.2 Ordering data.—Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Grade required (see 1.1).

(c) Odor requirement if other than as specified (see 3.1.2.2 and 3.1.3.2).

(d) Specific type of shipping containers required, if other than commercial-type containers (see 5.2).

6.3 Standard acid potassium phthalate for use in standardizing the sodium hydroxide solution (see 4.3.3.1.1), may be obtained from the Director, National Bureau of Standards, Washington 25, D. C.

6.4 The unit of purchase shall be the pound, net weight.

6.5 Transportation description.

6.5.1 *The transportation description applicable to this commodity is as follows:*

Glycerin, other than crude

(Released valuation not to exceed 50¢ per pound)

Carload minimum weight 30,000 pounds.
Truckload minimum weight 30,000 pounds.

Patent notice.—The furnishing by the Government of specifications, including drawings or other data in connection therewith, does not in any way grant the recipient of such material, or any other person, a license or right to manufacture, use, or sell any patented or unpatented invention that may in any way be related thereto.

7. DEPARTMENTAL REQUIREMENTS

7.1 The following specification and standard, of the issues in effect on date of invitation for bids, and special requirements, form a part of this specification for purchases made under this specification by the respective departments.

7.2 Army, Navy, and Air Force.

7.2.1 *Applicable specification and standard.*

Military Specification:

MIL–C–124—Containers (Cans, Pails and Drum) Metal (For Other Than Subsistence Items).

Military Standard:

MIL–STD–129—Marking for Shipment and Storage.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

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7.2.2 *Packaging.*

7.2.2.1 *Domestic shipment—immediate use.*—The glycerin shall be packaged as specified in 5.1.

7.2.2.2 *Domestic shipment and storage, or overseas shipment.*—The glycerin shall be furnished in 1-pint cans, 1-gallon cans, or in 5-gallon drums as specified (see 6.2). The 1-pint and 1-gallon cans shall conform to Military Specification MIL–C–124, type I, with screw-cap closures. The 5-gallon drums shall conform to Military Specification MIL–C–124, type V, with flexible spout closure, or with snap-on cap closure and rigid or flexible retractable metal spouts.

7.2.3 *Packing.*

7.2.3.1 *Domestic shipment—immediate use.*—The glycerin containers shall be packed for shipment as specified in 5.2.

7.2.3.2 *Domestic shipment and storage, or overseas shipment.*—The 1-pint and 1-gallon containers shall be packed for shipment in wood boxes in accordance with Appendix I to Military Specification MIL–C–124.

7.2.4 *Marking.*—The unit metal containers and the shipping containers shall be marked in accordance with Military Standard MIL–STD–129.

MILITARY INTERESTS:

Army—C E M O Q
Navy—A Or Sh
Air Force.

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